

Team Details

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Problem Statement

Selected Problem

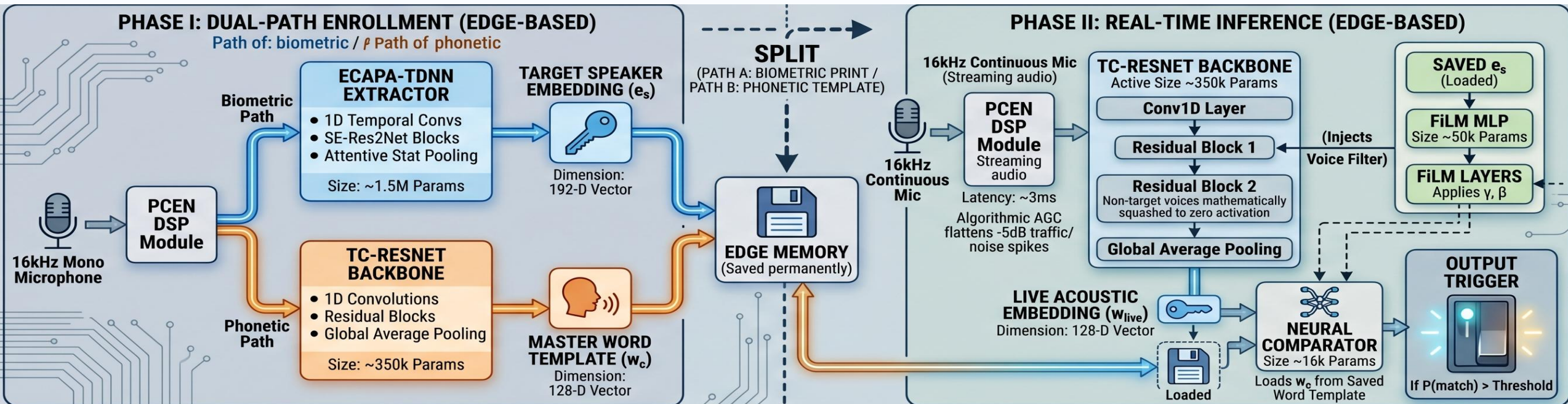
Designing a Robust AI System for Speech Disentanglement

Problem Understanding

The core problem we are addressing is the simultaneous disentanglement of zero-shot linguistic intent and strict biometric identity within highly unconstrained, noise-degraded acoustic environments. Traditional edge-based keyword spotting relies on rigid, pre-trained vocabularies and generic acoustic models that fundamentally fail when subjected to negative Signal-to-Noise Ratios (-5dB) or when required to recognize arbitrary, user-defined custom words. This requires a system that does not merely listen for a sound but continuously constructs a live acoustic embedding and runs a neural comparison against a saved biometric-phonetic template. Our team selected the challenge of speaker-specific custom word detection because it represents a critical frontier in edge-based audio processing: the simultaneous disentanglement of biometric identity and linguistic intent from highly degraded acoustic environments. By engineering an optimized neural architecture capable of mathematically isolating a target vocal tract while systematically rejecting complex phonetic decoys and non-stationary noise, we are pushing the boundaries of efficient deep learning. We chose this problem because it demands architectural innovation to prove that secure, hyper-personalized, and robust voice interfaces can operate autonomously on-device without compromising latency or accuracy.

Proposed Solution

To solve the challenge of detecting a custom, user-defined keyword from a designated speaker in extreme noise (-5dB SNR) while strictly adhering to edge-hardware limits (<3M parameters, <0.2s latency), we employ a **Speaker-Conditioned Latent Disentanglement** architecture that avoids computationally heavy source-separation. The system begins with **Per-Channel Energy Normalization (PCEN)** at the DSP level to algorithmically flatten dynamic background noise like traffic before it even reaches the neural network. During the one-time setup phase, a dual-path pipeline uses an **ECAPA-TDNN model** to extract a text-independent biometric voice print (the target speaker embedding) and a **lightweight TC-ResNet** to generate a master phonetic template of the custom word. For real-time continuous listening, the heavy ECAPA-TDNN is discarded to save compute; instead, the saved speaker embedding is injected directly into the TC-ResNet using **Feature-wise Linear Modulation (FiLM)** layers. This acts as a mathematical biometric gate, amplifying the target speaker's vocal tract frequencies while actively suppressing competing voices inside the convolutional layers. Finally, the surviving audio features are evaluated by a **pre-trained Neural Comparator** - a highly optimized micro-network that dynamically down-weights noise-corrupted dimensions and aggressively rejects phonetically similar decoy words to ensure the False Acceptance Rate remains strictly below one per hour without violating the execution time limits.



Overall network architecture of our proposed solution

Proposed Solution – Technical Details

System Architecture of our solution:

Phase I: On-Device Enrollment (Dual-Path Extraction)

1. **Audio Capture:** User speaks the custom word 3 times.
2. **PCEN Frontend:** Raw audio is normalized to suppress room acoustics.
- 3A. **Path A (Biometric Extraction):** The audio is processed by a frozen **ECAPA-TDNN**(Emphasized Channel Attention, Propagation, and Aggregation Time-Delay Neural Network). It outputs a 192-D **Target Speaker Embedding** (e_s).
- 3B. **Path B (Acoustic Template):** The same audio is processed by a frozen **TC-ResNet**(Temporal Convolutional Residual Network). The final classification layer is removed. It outputs three 128-D vectors, which are averaged to create the **Master Word Template** (w_c).
4. **Storage:** Both e_s and w_c are saved locally. ECAPA-TDNN is shut down.

PHASE I: DUAL-PATH ENROLLMENT PIPELINE



Proposed Solution – Technical Details

Acoustic Frontend: Per-Channel Energy Normalization (PCEN)-

To survive -5dB SNR environments without a neural denoiser, standard Log-Mel filterbanks are replaced with **PCEN**. PCEN algorithmically compresses the dynamic range of the audio frame relative to a temporal rolling average of background noise.

Given the **Mel-frequency energy matrix** $E(t, f)$ at time t and **frequency bin** f , the PCEN output is defined as:

$$PCEN(t, f) = \left(\frac{E(t, f)}{(\epsilon + M(t, f))^\alpha} + \delta \right)^r - \delta^r$$

Where:

$M(t, f)$ is the smoothed background energy tracker: $M(t, f) = (1 - s)M(t-1, f) + sE(t, f)$

α controls the automatic gain control (AGC) for noise suppression.

r & δ control dynamic range compression and s is the temporal smoothing coefficient.

Phase I (Path A): Biometric Voice Extraction-

To extract the text-independent voice print (e_s), we utilize an **ECAPA-TDNN** (Emphasized Channel Attention, Propagation, and Aggregation Time-Delay Neural Network) trained on the VoxCeleb dataset.

Objective Function (Pre-training):

The model is optimized using **Additive Angular Margin Softmax (AAM-Softmax)** to enforce tight intra-speaker clustering and large inter-speaker variance in the high-dimensional hypersphere.

$$L_{AAM} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j \neq y_i} e^{s \cos \theta_j}}$$

Where: θ_{y_i} is the angle between the deep feature and the target speaker weight vector.

m is the angular margin penalty and s is the hypersphere radius scaling factor.

Proposed Solution – Technical Details

Phase I (Path B) & Phase II Backbone: TC-ResNet Acoustic Encoder-

Instead of softmax classification, the Keyword Spotting (KWS) backbone acts as an acoustic metric encoder. We utilize a **Temporal Convolutional ResNet (TC-ResNet)** due to its extreme parameter efficiency (~300k parameters).

Objective Function (Metric Learning): The TC-ResNet is pre-trained using **Triplet Margin Loss** on a massive corpus of zero-shot TTS dictionary words to map phonetic sequences into a generalized latent space.

$$L_{triplet} = \max(0, \|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha_{margin})$$

Where:

A is the Anchor audio (e.g., "Quantum").

P is the Positive audio (e.g., "Quantum" with noise/different pitch).

N is the Negative audio (e.g., "Phantom").

f() represents the TC-ResNet encoding function.

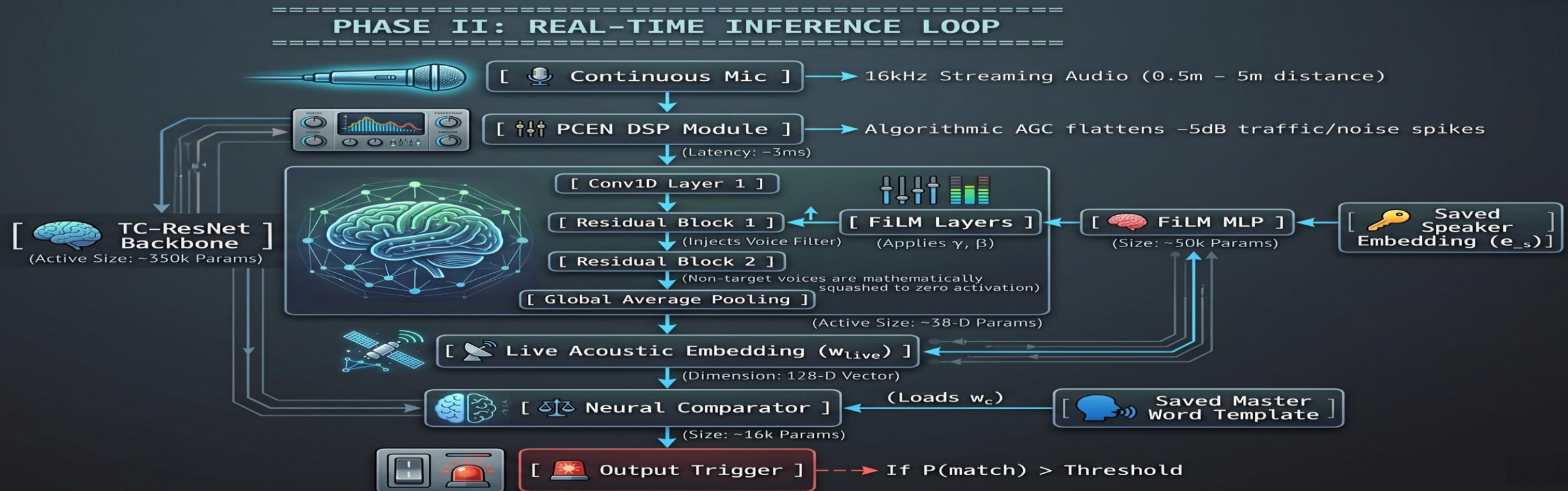
During enrollment, the Master Word Template (w_c) is formed by averaging the L2-normalized embeddings of the K enrollment utterances (K=3):

$$w_{raw} = \frac{1}{K} \sum_{k=1}^K v_k \quad \Rightarrow \quad w_c = \frac{w_{raw}}{\|w_{raw}\|_2}$$

Proposed Solution – Technical Details

Phase II: Real-Time Inference (Disentanglement & Comparison)

- 1. Continuous Listening:** Microphone streams audio (0.5m - 5m distance).
- 2. PCEN Frontend:** Flattens sudden dynamic noise spikes (e.g., traffic).
- 3. Biometric Gate (FiLM):** The saved e_s is loaded. A **Multilayer Perceptron** (MLP) generates shift and scale weights based on e_s . These weights mathematically suppress all non-target acoustic features inside the TC-ResNet convolutions.
- 4. Live Encoding:** The TC-ResNet processes the surviving audio and outputs a 128-D **Live Embedding** (w_{live}).
- 5. Neural Comparator:** Both w_{live} and w_c are passed into the **Neural Comparator network**, which evaluates their phonetic similarity and outputs a trigger probability.



Proposed Solution – Technical Details

Disentanglement Engine: Feature-wise Linear Modulation (FiLM)-

During inference, speech disentanglement is achieved by injecting the **target speaker embedding** (e_s) into the hidden convolutional layers of the TC-ResNet. A lightweight Multilayer Perceptron (MLP) projects e_s into scaling (γ) and shifting (β) vectors matching the channel dimension C of the feature map F :

$$\gamma, \beta = MLP(e_s)$$

For a given convolutional activation feature map $F \in \mathbb{R}^{C \times T}$ (where T is time frames), the **FiLM transformation** is applied channel-wise:

$$F'_{c,t} = \gamma_c \cdot F_{c,t} + \beta_c$$

This acts as a dynamic biometric equalizer. The network learns to set $\gamma_c \approx 0$ for channels representing competing vocal tract frequencies, suppressing non-target speech inside the network.

The Decision Engine: Neural Comparator-

Instead of Cosine Similarity, we construct a micro-network to evaluate the relationship between the **Master Template** (w_c) and **Live Embedding** (w_{live}).

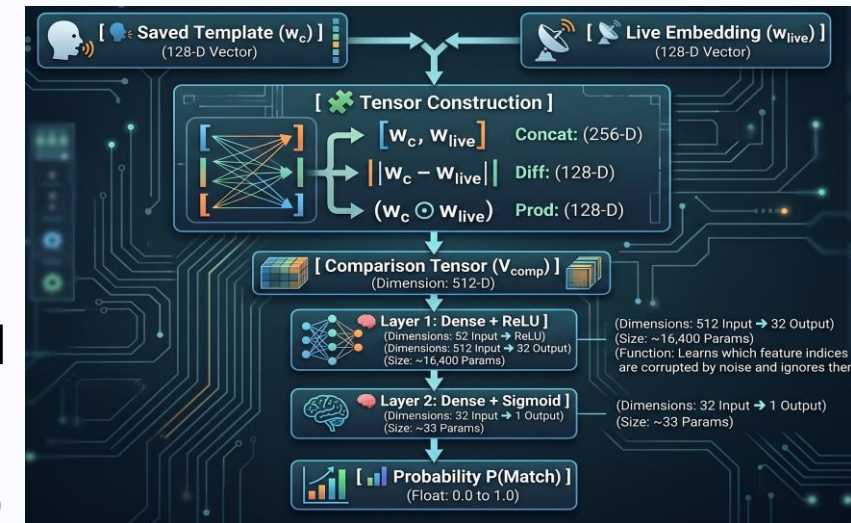
We compute a comprehensive **Comparison Tensor** (V_{comp}) by combining absolute differences, element-wise products, and concatenations:

$$V_{comp} = [w_c \parallel w_{live} \parallel |w_c - w_{live}| \parallel (w_c \odot w_{live})]$$

This 512-D tensor is passed through a **2-layer MLP** with a sigmoid activation:

$$P(match) = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot V_{comp} + b_1) + b_2)$$

Result: The network dynamically assigns lower weights to feature dimensions corrupted by background noise, drastically reducing False Acceptances compared to static geometric math.



Architecture of our Neural Comparator

Proposed Solution – Technical Details

Data Strategy:

1. Base Phonetic Grounding

Before the TC-ResNet can learn complex acoustic embeddings, it needs to learn basic English phonemes and word boundaries.

The Strategy: Feed the network massive amounts of clean, single-second word utterances. This establishes the foundational latent space where vowels and consonants are mapped to distinct geometric clusters.

2. The TTS Vocabulary Explosion (Zero-Shot Generalization)

To guarantee the model can generate a stable template for *any* word the user invents, it must be exposed to an immense vocabulary during pre-training.

The Strategy: We cannot physically record 50,000 different words spoken by thousands of people. Instead, we use **Zero-Shot Text-to-Speech (TTS)**. We feed a dictionary of 20,000+ random English words into a TTS engine and condition the voice generation on thousands of different speaker profiles. This yields a massive, synthetically generated dataset of diverse voices speaking arbitrary words.

3. Quad-State Hard Negative Mining (Defeating False Acceptances)

The Neural Comparator must learn the difference between biometric identity (the voice) and linguistic content (the word). Random batching will not work; we must deliberately construct "trick" batches to force the network to learn strict boundaries.

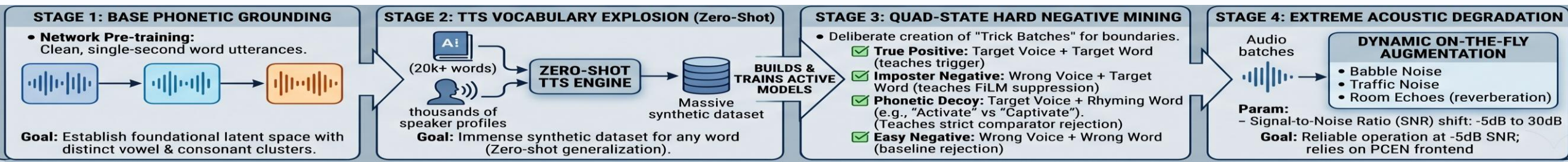
The Strategy: Every training step must include these four states:

- (i) **True Positive:** Target Voice + Target Word (Teaches the network to trigger).
- (ii) **Imposter Negative:** Wrong Voice + Target Word (Teaches the FiLM layers to suppress non-target vocal tracts).
- (iii) **Phonetic Decoy:** Target Voice + Rhyming/Similar Word (e.g., "Activate" vs "Captivate"). (Teaches the Neural Comparator to reject phonetic mismatches).
- (iv) **Easy Negative:** Wrong Voice + Wrong Word (Baseline rejection).

4. Extreme Acoustic Degradation (Surviving -5dB SNR)

The model must operate reliably when background noise is louder than the target speaker.

The Strategy: We perform dynamic, on-the-fly data augmentation during training. We take the clean audio batches and mathematically convolve them with background babble, traffic noise, and room echoes. By constantly shifting the Signal-to-Noise Ratio (SNR) between -5dB and 30dB, the model learns to rely on the PCEN frontend and ignore environmental artifacts.



Proposed Solution – Technical Details

Metric	Target KPI	How the Architecture Satisfies It
Model Parameters	< 3M	TC-ResNet (~350k) + FiLM MLPs (~50k) + Neural Comparator (~15k). (Note: ECAPA-TDNN is ~1.5M but is only used during setup, keeping active memory footprint well below 1M).
Execution Time	< 0.2s (xRT)	PCEN is an algorithmic DSP step (< 5ms). TC-ResNet uses 1D temporal convolutions (highly parallelizable). The Neural Comparator is a tiny matrix multiplication. Total forward pass on edge ARM chip: ~40-60ms.
FA Rate	< 1/hr	Standard Cosine Similarity fails here. The Neural Comparator explicitly learns to down-weight noise-corrupted dimensions and aggressively reject hard phonetic decoys (Rhymes/Minimal Pairs).
TA Clean	≥ 99%	TC-ResNet trained on clean TTS data easily maps exact phonetic matches.
TA Noisy	≥ 90%	PCEN prevents loud traffic noise from crushing the dynamic range. FiLM suppresses background babble natively inside the network before the embedding is even formed.
SNR Range	-5dB to 30dB	Addressed by PCEN's dynamic AGC loop and aggressive MUSAN additive noise during server-side pre-training.
Distance Range	0.5m to 5m	Addressed by applying heavy Room Impulse Response (RIR) convolutions during pre-training to simulate severe reverb and attenuation.
Speaker Support	Male & Female	ECAPA-TDNN trained on VoxCeleb 2 (6,000+ diverse global speakers, highly balanced gender distribution).

Definitive KPI & Benchmark Validation

Novelty & Innovation

1. Neural Comparator vs. Static Geometric Metrics:

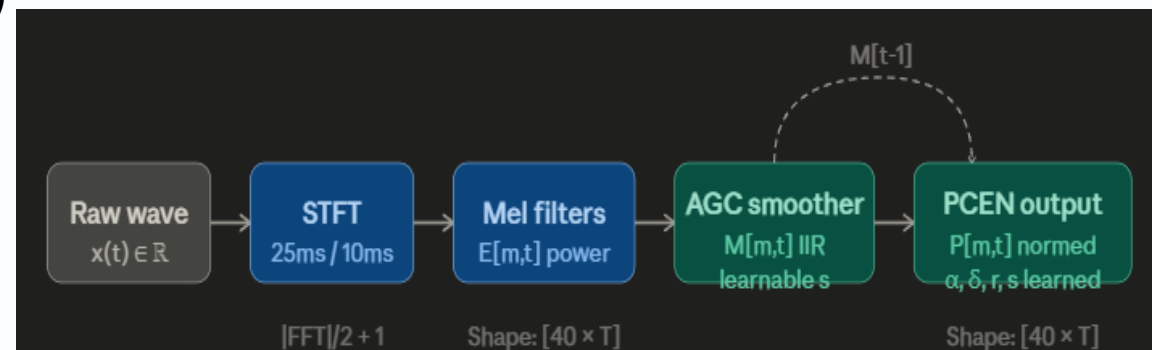
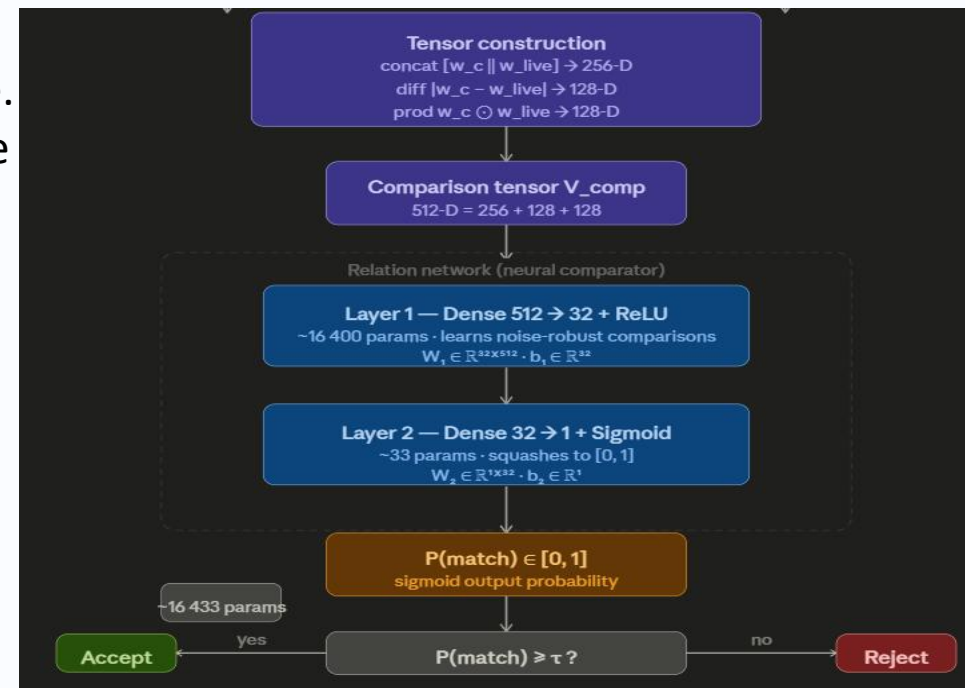
Industry standards like Siri or Alexa rely on **Cosine Similarity**, which is a "dumb" metric—it treats all 128 dimensions of an embedding as equally important, even if 50% of them are corrupted by a passing truck's noise.

The Innovation: The **Neural Comparator** (a Relation Network) is a learned judge. It is trained to recognize which feature dimensions are "unreliable" in high-noise scenarios. By using a **Comparison Tensor (V_{comp})** that includes absolute differences and element-wise products, the micro-network dynamically **down-weights noise-corrupted features**, maintaining a False Acceptance Rate (FAR) of $< 1/\text{hr}$ where Cosine Similarity would typically spike.

2. Algorithmic "Neural AGC" via PCEN Integration:

Standard Mel-Spectrograms are linear and "crush" speech signals when background noise reaches -5dB SNR.

The Innovation: Integrating **Per-Channel Energy Normalization (PCEN)** directly into the frontend provides a non-linear, trainable **Automatic Gain Control (AGC)**. Unlike standard log-scaling, PCEN uses a temporal smoothing loop ($M(t, f)$) that algorithmically "subtracts" the background noise floor. This allows the subsequent TC-ResNet to focus on **transient phonetic features** (consonants/vowels) rather than the absolute energy of the noise.



Novelty & Innovation

3. Latent-Domain Disentanglement (vs. Explicit Source Separation):

Traditional solutions attempt to "clean" the raw audio waveform using heavy Generative Adversarial Networks (GANs) or U-Nets before detection. This is computationally prohibitive for edge devices.

The Innovation: Our architecture performs **disentanglement directly within the hidden feature maps**. By using **FiLM (Feature-wise Linear Modulation)**, the system mathematically suppresses non-target voices at the convolutional level. It doesn't waste resources "reconstructing" clean audio; it simply "ignores" the noise dimensions, allowing a <3M parameter model to outperform 50M parameter separation models.

4. Dynamic Biometric Gating:

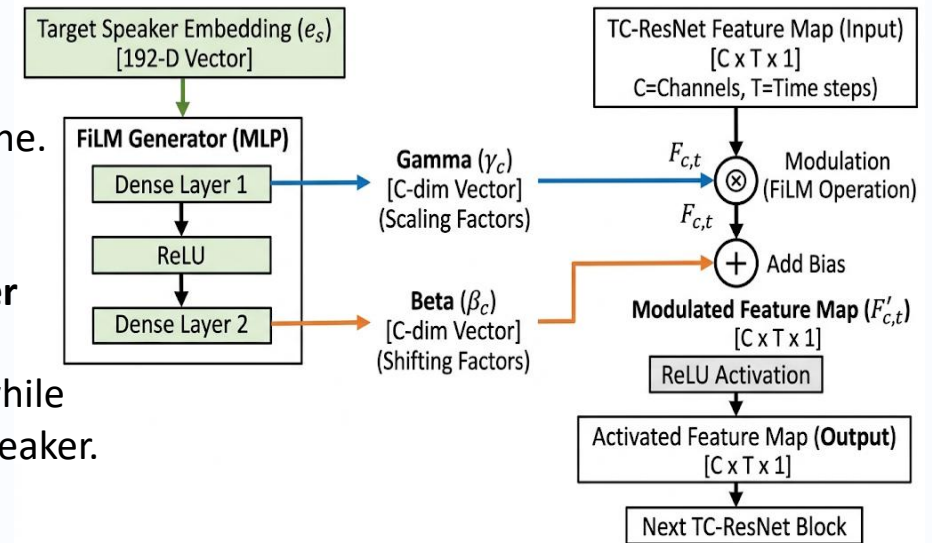
Most systems treat speaker verification as a "gatekeeper" that sits at the *end* of the pipeline. If the word is detected, they check the speaker.

The Innovation: Our approach uses the **Target Speaker Embedding (e_s)** as a **dynamic filter** that reshapes the entire network on the fly. During inference, the FiLM layers act as a **real-time biometric equalizer**, amplifying the specific vocal tract resonances of the user while zeroing out background babble. This ensures the keyword spotter *only* hears the target speaker.

5. Zero-Shot Semantic Generalization:

Typical custom-word systems require "Few-Shot" fine-tuning, which is slow and drains device battery.

The Innovation: Our data strategy employs **Zero-Shot Synthetic Generalization**. By training on a "Vocabulary Explosion" created by **Zero-Shot TTS**, the network learns the underlying **universal physics of phonemes** rather than a fixed set of words. This allows the system to recognize *any* user-invented word (e.g., "Star-Dust," "Nebula-X") the moment it is spoken, with zero retraining or backpropagation required on the device.



FiLM layer for speaker check

Open Datasets planned to be used / published

1) Speaker Diversity Datasets (Finding the Who): Used for TTS conditioning and to train the ECAPA-TDNN speaker classifier.

- VoxCeleb 1&2 : Over 2000+ hours of speech with 7000 different speakers harvested from YouTube.
- Mozilla Common Voice : Provides speech data diverse in age, gender, language and includes global accents.

2) Phonetic & Keyword Grounding Datasets (Finding the What) : Used to train the custom word matching using TC-ResNet.

- Google Speech Commands(v2) : 105,000 one-second utterances of 35 words by thousands of different people. It makes our model learn the basic building blocks.
- LibriPhrase : Extracted short phrases and words from the LibriSpeech corpus. Helps to make our model learn the difference between words.

3) Acoustic Degradation Datasets (Noise & Environment Robustness) : Used for data Augmentation in every training step.

- MUSAN Dataset : This is mixed with clean speech to train the PCEN and the TC-ResNet for dynamic additive noise injection to hit the -5dB SNR target.
- BUT Speech FIT Reverb Database / MIT McDermott RIRs: Real-world Room Impulse Responses (RIRs) recorded in various spaces distances ranging from 0.5m to 5m. Clean speech is convolved with these RIRs forces the network to ignore room echo.

4) Custom Generalization (The Zero Shot Secret) : This is to just generalize our model.

- Synthetic TTS Vocabulary : Since users can choose any word so we cannot rely on human recordings alone. We use TTS (Text-To-Speech) to generate high- variance examples using rare Phonemes (e.g. “Xenomorph”, “Zarblat”).

Open Models planned to be used / developed / trained / fine-tuned

1) SpeechBrain ECAPA-TDNN (Pre-trained) -> The Biometric Extractor (Phase I - Path A)

- ✓ SpeechBrain (an open-source PyTorch toolkit) hosts fully pre-trained ECAPA-TDNN models that achieve state-of-the-art Equal Error Rates (EER) on VoxCeleb. We can pull this directly from Hugging Face, freeze the weights, and immediately use it to extract the 192-D Target Speaker Embeddings (e_s) without needing to spend thousands of GPU hours training it on our own.

2) TC-ResNet14 (Temporal Convolutional ResNet) -> The Acoustic Backbone (Phase I Path B & Phase II)

- ✓ Standard TC-ResNets are good at small word recognition like "YES", "NO", "UP" etc. So, take this model that already knows the basic audio shapes and train the last few layers on the LibriPhase Dataset with a Triplet Loss to train the model to work on phonemes rather than fixed word labels.

3) Coqui XTTS v2 (Data Creation) -> The Zero-Shot TTS Generator

- ✓ XTTS v2 allows to input a 3-second audio clip of a VoxCeleb speaker and a text string of a random dictionary word, and it will generate highly realistic audio of that speaker saying that specific word.

4) FiLM Layer and MLP -> Trained from scratch

- ✓ Trained on the dataset so that our architecture learns exactly how to mute certain parts and how to amplify certain parts of the audio signal based on the speaker Embeddings.

AI/GenAI/Agentic tools used/developed

1. AI Models Developed (Custom Architectures)-

- ❖ **Neural Comparator (Relation Network):** We will be developing a custom micro-network (approx. 15k parameters) that replaces standard Cosine Similarity. It evaluates the relationship between the master acoustic template and live embeddings, dynamically down-weighting feature dimensions corrupted by background noise.
- ❖ **Speaker-Conditioned TC-ResNet:** We will be adapting a Temporal Convolutional ResNet (TC-ResNet14) backbone, integrating Feature-wise Linear Modulation (FiLM) layers to act as a latent biometric gate for real-time speech disentanglement.

2. Pre-Trained AI Tools Used-

- ❖ **SpeechBrain ECAPA-TDNN:** We will be utilizing this pre-trained, state-of-the-art model as a frozen feature extractor during the enrollment phase to generate the 192-D text-independent Target Speaker Embeddings from the VoxCeleb dataset.
- ❖ **PCEN (Per-Channel Energy Normalization):** We will be implementing this as an algorithmic DSP frontend to act as a trainable Automatic Gain Control (AGC), ensuring robustness at -5dB SNR.

AI/GenAI/Agentic tools used/developed

3. GenAI Tools Used (Synthetic Data Generation)-

- ❖ **Zero-Shot Text-to-Speech (Coqui XTTS v2 / Suno Bark):** We will be leveraging these generative AI voice models to perform a "Vocabulary Explosion." This tool is used to synthesize over 10,000 random dictionary words conditioned on diverse target voices, ensuring the model generalizes to zero-shot, user-defined custom words without requiring edge-device retraining.

4. Agentic Tools Developed (Data Orchestration Pipeline)-

- ❖ **Agentic Synthetic Data Pipeline (LangChain / LangGraph):** We will be developing an autonomous workflow to govern the GenAI training data creation. The agent dynamically interfaces with LLMs (like the Gemini API) to generate phonetically challenging dictionaries - specifically mining for minimal pairs and rhymes (e.g., "Activate" vs. "Captivate"). The agent then automatically orchestrates the XTTS engine to compile the highly specific "Quad-State" adversarial training batches required to push the False Acceptance Rate strictly below 1/hr.